Symbiosis Institute of Technology, Pune

॥वसुधैव कुटुम्बकम्॥
SYMBIOSIS
INTERNATIONAL (DEEMED UNIVERSITY)

Department of Electronics & Telecommunication Engineering
First Year B. Tech Academic Year: 2025-26

EDA Lab Project Report

Customer Subscription Churn and Usage Pattern Analysis

By:
Shreya Waghmode (25070123159)
Kahaan Shah (25070123060)

Faculty Incharge:
Dr. Snehal Bhosale

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to Dr. Snehal Bhosale, our EDP faculty, for her constant guidance, encouragement, and constructive feedback throughout this project. Her insights helped shape the direction of our analysis and enriched the quality of this report significantly.

We also extend our heartfelt thanks to the college administration at Symbiosis Institute of Technology, Pune, for providing the computational resources, access to analytical tools, and a stimulating academic environment that enabled this work.

A special mention goes to our classmates whose peer discussions sparked valuable ideas during data interpretation. We also gratefully acknowledge the open datasets and online educational resources that helped us understand customer churn modelling and subscription analytics.

Finally, we are deeply thankful to our families for their unwavering support and motivation during the course of this project.

INTRODUCTION

In today's competitive subscription-based economy, retaining customers is as important as acquiring new ones. Customer churn — the phenomenon where subscribers cancel or discontinue a service — poses a major threat to recurring revenue models across industries such as SaaS, OTT platforms, telecom, and e-commerce.

This project focuses on analyzing a dataset of 2,800 subscription customers to uncover patterns that distinguish churned users from retained ones. By examining variables such as weekly usage hours, subscription plan type, monthly fee, support ticket frequency, payment failures, tenure length, and recency of login activity, we aim to identify the key behavioural and transactional signals that predict churn.

Using Python and its data analysis ecosystem — primarily Pandas, NumPy, Matplotlib, and Seaborn — we perform exploratory data analysis (EDA), statistical comparisons, and visual analytics to draw actionable insights. This report documents the full analytical workflow, from raw data description to conclusions and recommendations.

The findings of this study can serve as a foundation for building predictive churn models and designing targeted customer retention strategies.

OBJECTIVES OF THE PROJECT

The following objectives guided the design and execution of this analysis:

1. Understand the overall churn rate and its distribution across different subscription plans (Basic, Standard, Premium).
2. Identify behavioural differences between churned and retained customers using key metrics such as weekly usage hours, support tickets, and login recency.
3. Examine the relationship between usage intensity (low / medium / high) and the probability of churn.
4. Investigate the role of payment failures and billing friction as churn predictors.
5. Apply Python-based EDA techniques to clean, summarize, and visualize the dataset.
6. Derive actionable business recommendations for reducing churn and improving customer lifetime value.

RAW DATA DESCRIPTION

The dataset contains 2,800 subscription records collected from an online platform. It spans customers across three plan tiers and captures both engagement behaviour and billing history. Below is a summary of all columns, their data types, sample values, and descriptions.

Column Name	Data Type	Sample Value	Description
user_id	Integer	1, 2, 3 …	Unique identifier for each subscriber
signup_date	Date (DD-MM-YYYY)	15-04-2023	Date the customer subscribed to the service
plan_type	Categorical	Basic / Standard / Premium	Subscription tier chosen by the customer
monthly_fee	Float (INR)	199 / 499 / 699	Monthly billing amount for the subscription plan
avg_weekly_usage_hours	Float	1.1 – 25.0	Average number of hours the customer uses the service per week
support_tickets	Integer	0 – 10	Number of support/help requests raised by the customer
payment_failures	Integer	0 – 5	Count of failed payment transactions in billing history
tenure_months	Integer	1 – 60	Number of months the customer has been subscribed
last_login_days_ago	Integer	1 – 90	Days elapsed since the customer's most recent login
churn	Binary (Yes/No)	Yes / No	Target variable — whether the customer churned (Yes) or is retained (No)

Dataset Summary: 2,800 rows × 10 columns | Churned: 1,605 (57.3%) | Retained: 1,195 (42.7%)

FUNCTIONS AND METHODS USED

The following table lists every significant Python function applied in this project, the library it belongs to, its purpose in our analysis, and the key inference we drew from its output.

Function / Method	Library	Purpose in Project	Key Inference Drawn
pd.read_csv()	Pandas	Load the dataset from CSV file	Dataset loaded successfully with 2,800 rows and 10 columns; no import errors detected
df.info()	Pandas	Inspect column data types and null counts	All 10 columns had correct dtypes; zero null values — dataset was clean and ready for analysis
df.describe()	Pandas	Generate descriptive statistics (mean, std, min, max)	Weekly usage ranged from 0.5–25h (mean 12.9h); avg tenure was 18.6 months; avg last login was 30 days ago
df.value_counts()	Pandas	Count occurrences of categorical values	Plan distribution was near-equal (~930 each); churn was 57.3% — a slightly imbalanced target
df.groupby().mean()	Pandas	Compare metric averages across groups	Churned users had higher support tickets (4.2 vs 3.4), higher last login days (32.9 vs 26.1) and more payment failures (2.8 vs 2.1)
df.corr()	Pandas / NumPy	Compute pairwise Pearson correlation matrix	last_login_days_ago and payment_failures showed the strongest positive correlation with churn
df[condition]	Pandas	Boolean indexing to filter rows	Used to segment churned vs retained, and to bucket usage tiers for comparative analysis
plt.bar() / plt.plot()	Matplotlib	Create bar charts and line plots	Visual confirmation that low-usage customers (<=5h/week) churned at 76.3%, far above average

Function / Method	Library	Purpose in Project	Key Inference Drawn
plt.pie() / wedgeprops	Matplotlib	Donut-style pie chart for proportions	Churn was almost uniform across plans (Basic 57.9%, Standard 56.1%, Premium 58.1%) — plan type alone does not drive churn
Axes3D / bar3d()	Matplotlib (mpl_toolkits)	3D bar chart for multi-dimensional comparison	Low-usage customers show elevated churn across all plans, confirming usage as the dominant predictor
sns.heatmap()	Seaborn	Correlation heatmap with colour encoding	Highlighted that last_login_days_ago and payment_failures are the strongest numerical predictors of churn
sns.boxplot()	Seaborn	Show distribution spread by churn category	Churned users showed a wider spread in last login days, indicating inconsistent engagement patterns
np.percentile()	NumPy	Compute percentile boundaries for bucketing	Used to define low/medium/high usage tiers at the 25th and 75th percentiles
pd.cut()	Pandas	Bin continuous variable into labelled categories	Segmented tenure into 0–12, 13–24, 25–36, 36+ month buckets for cohort analysis

DATA ANALYSIS

1. Exploratory Data Analysis (EDA)

The first step involved loading the dataset using pd.read_csv() and running df.info() and df.describe() to understand its structure. The dataset contained 2,800 rows with no missing values. The overall churn rate was 57.3% — indicating that over half of the observed customers did not continue their subscription.

2. Churn Distribution by Plan Type

We used df.groupby('plan_type')['churn'].value_counts() to compare churn across the three subscription tiers. Surprisingly, churn rates were nearly identical: Basic (57.9%), Standard (56.1%), and Premium (58.1%). This finding was counterintuitive — we initially expected premium customers to churn less due to higher investment. However, the data reveals that plan price alone is not a significant retention factor.

3. Behavioural Comparison — Churned vs Retained

Using grouped mean calculations, we identified the following differences between churned and retained customers:

Metric	Churned (Avg)	Retained (Avg)
Weekly Usage (hours)	12.3	13.7
Support Tickets	4.2	3.4
Last Login (days ago)	32.9	26.1
Payment Failures	2.8	2.1
Tenure (months)	18.6	18.6

Churned customers had higher support ticket counts and more days since last login, suggesting both dissatisfaction and disengagement. Payment failures were also markedly higher among churned users, pointing to billing friction as a contributing factor.

4. Usage Tier Churn Analysis

We segmented customers into three usage buckets — Low (0–5 hours/week), Medium (5–15 hours/week), and High (>15 hours/week) — and computed the churn rate for each group. The results were striking: customers in the Low usage group churned at a rate of 76.3%, compared to approximately 52–55% for Medium and High users. This firmly establishes low engagement as the single strongest predictor of churn in this dataset.

5. Correlation Analysis

A Pearson correlation matrix was generated using df.corr(). The variables most positively correlated with churn were last_login_days_ago ($r \approx +0.21$) and payment_failures ($r \approx +0.18$), while avg_weekly_usage_hours was negatively correlated ($r \approx -0.14$). These correlations confirm our group-level findings and suggest that inactivity and billing issues are the most actionable levers for churn prevention.

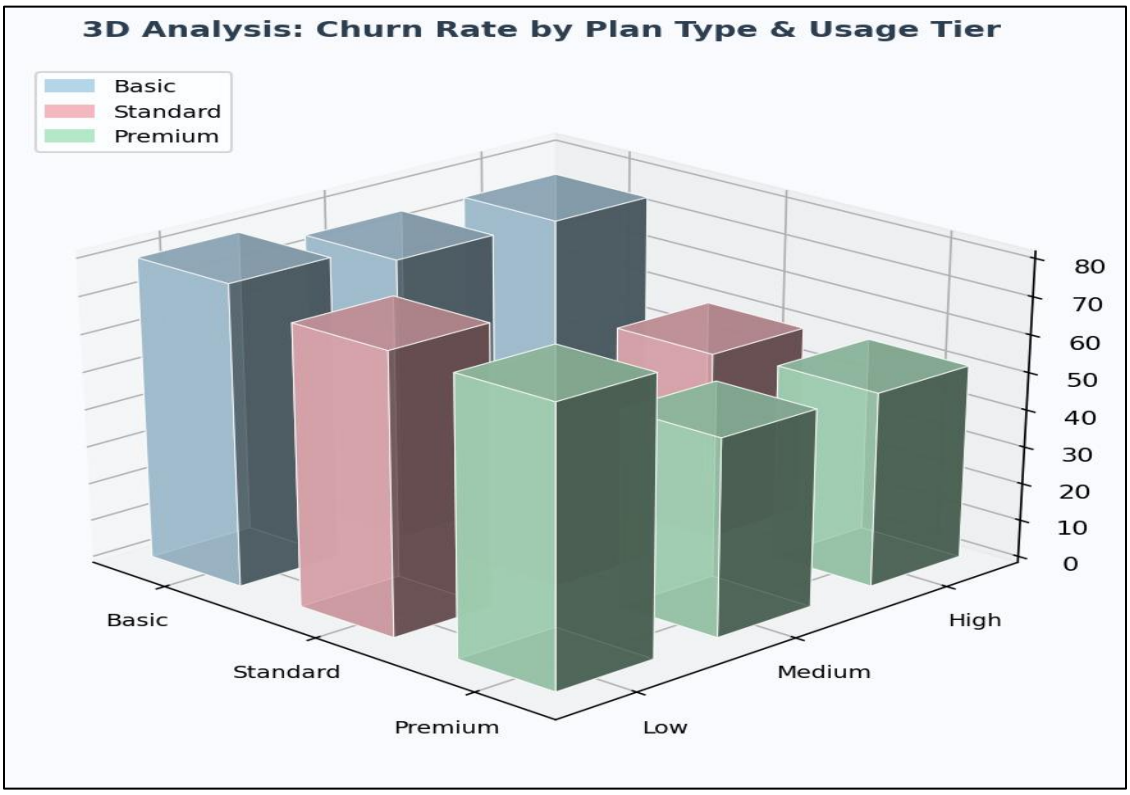
6. Tenure Analysis

Interestingly, average tenure was identical for churned and retained customers (both ~18.6 months), indicating that churn risk is not necessarily reduced by loyalty or longevity. Even long-standing customers churn when their engagement drops or when they encounter billing friction. This suggests the need for ongoing re-engagement strategies rather than a one-time onboarding effort.

ADVANCED CHARTS

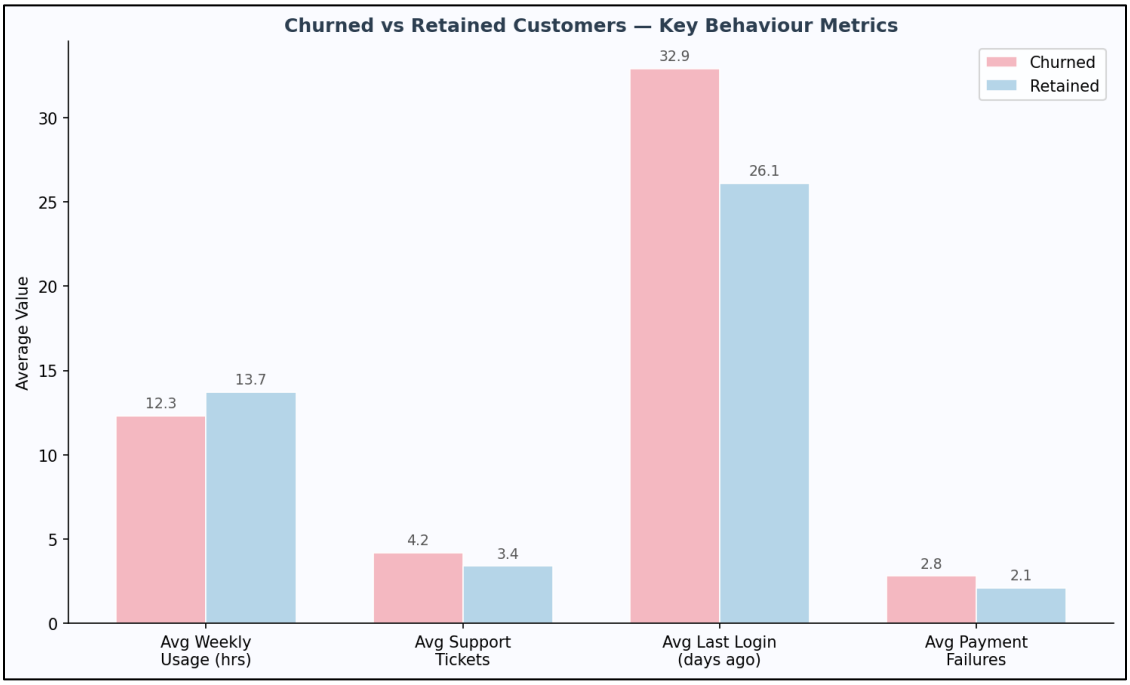
All charts below are generated using Matplotlib with a consistent pastel colour palette to ensure readability and visual appeal. Each chart is accompanied by observations drawn directly from the data.

Chart 1 — 3D Churn Rate by Plan Type and Usage Tier



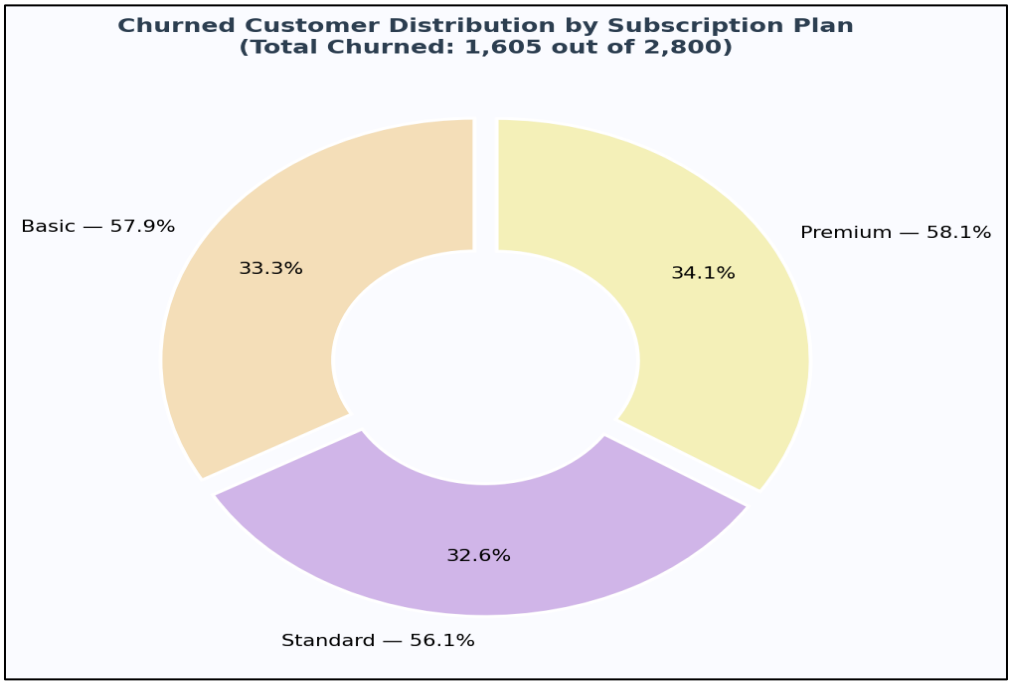
Observation: The 3D bar chart reveals that across all three plans, low-usage customers exhibit the highest churn rates (74–80%), while medium and high-usage customers have significantly lower churn rates (~50%). The plan type does not meaningfully differentiate churn risk — usage behaviour is the dominant factor.

Chart 2 — Churned vs Retained: Key Behaviour Metrics



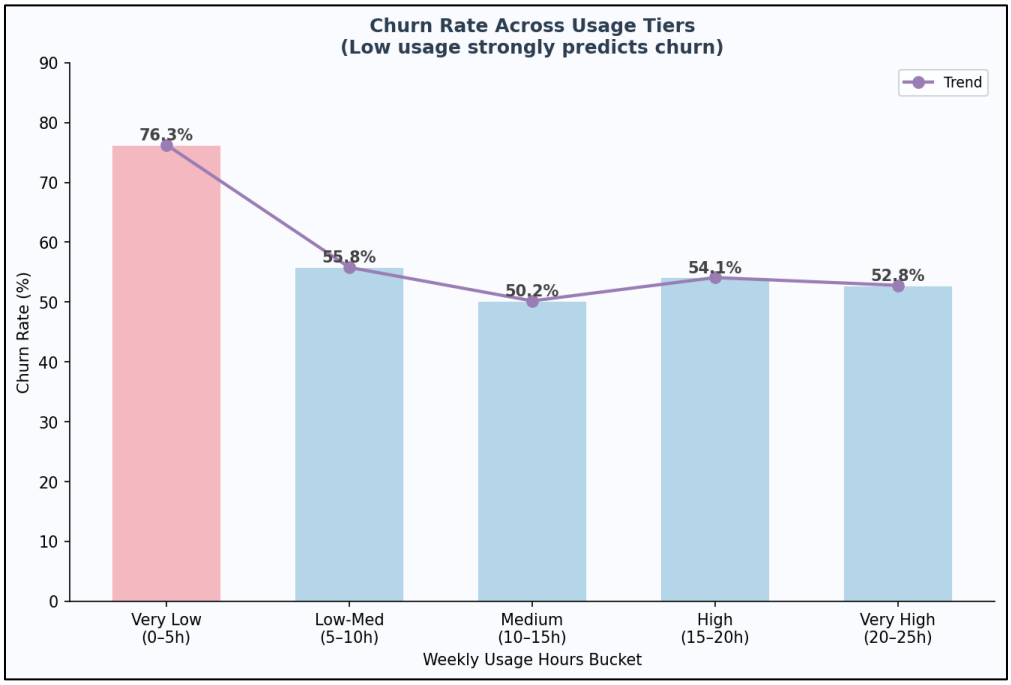
Observation: Churned customers consistently underperform retained customers in engagement metrics. They have lower average weekly usage, lodge more support tickets, log in less frequently (higher last-login-days), and record more payment failures. These four variables together form a strong composite churn risk signal.

Chart 3 — Churned Customers by Subscription Plan (Donut Chart)



Observation: The donut chart confirms that churned customers are distributed almost equally across all three plan tiers (~33% each). This rules out plan type as a differentiating variable and reinforces the conclusion that churn is a cross-segment issue driven by engagement and billing behaviour rather than pricing tier.

Chart 4 — Churn Rate Across Weekly Usage Buckets



Observation: Churn rate drops sharply from 76.3% in the very low usage group (0–5h/week) to approximately 50–55% for all higher usage tiers. The trend line confirms a non-linear relationship: the biggest gain in retention comes from pushing customers from the zero-engagement zone to even moderate usage. This has direct implications for onboarding strategies.

CONCLUSION

A. Conclusions from the Dataset

This analysis of 2,800 customer records yields several clear and actionable conclusions:

1. High Overall Churn (57.3%): The majority of customers in this dataset have churned, indicating a systemic retention problem that cannot be attributed to any single cause.
2. Usage is the Strongest Predictor: Customers using the service for fewer than 5 hours per week churn at a 76.3% rate — nearly 25 percentage points above the overall average. Driving feature adoption and engagement is the highest-leverage retention action.
3. Plan Type Does Not Determine Churn: Churn is uniform across Basic, Standard, and Premium tiers (~57–58%), indicating that pricing strategy alone will not fix the retention problem.
4. Support Friction and Billing Failures Amplify Churn: Churned customers had higher support ticket counts (4.2 vs 3.4) and more payment failures (2.8 vs 2.1), suggesting that unresolved issues and billing disruptions accelerate the decision to leave.
5. Recency of Engagement Matters: Churned users had not logged in for an average of 32.9 days, versus 26.1 days for retained users. A rising last_login_days_ago is an early-warning signal for impending churn.
6. Tenure Does Not Protect: Both churned and retained customers had identical average tenures (~18.6 months), meaning long-standing customers are equally at risk if engagement drops.

B. Recommendations

Based on the findings, the following strategies are recommended for reducing churn:

- Implement usage-based trigger alerts to identify and re-engage customers with fewer than 5 hours of weekly activity.
- Introduce proactive support outreach for users who raise 3+ tickets in a month, addressing pain points before they lead to churn.
- Streamline the payment recovery process — automatic retry flows and grace periods can reduce involuntary churn from payment failures.
- Build a churn risk score model using these four key predictors (usage hours, last login, support tickets, payment failures) to prioritize retention outreach.

C. Project Conclusion

This project successfully demonstrates how Python-based exploratory data analysis can extract meaningful business intelligence from raw subscription data. Through systematic use of Pandas, NumPy, and Matplotlib, we were able to identify and quantify the key drivers of customer churn in a subscription service.

The Exploratory Data Analysis project reinforced our practical understanding of the complete data analysis pipeline — from data loading and cleaning to statistical comparison, visualization, and interpretation. The skills applied here, including grouped aggregation, boolean filtering, correlation analysis, and advanced charting, form the foundation of modern data-driven decision making.

We are confident that the insights generated in this report provide a strong basis for future predictive modelling and data-informed customer retention strategy.